

Summary: *How Puzzling Is the Forward Premium Puzzle? A Meta-Analysis*

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Forthcoming in the *European Economic Review*

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1 Paper Overview

1.1 Introduction & Context

A foundational prediction in financial economics is that, under risk neutrality and rational expectations, a currency's forward exchange rate should be an unbiased predictor of the future spot rate. In the standard regression framework, this means the slope coefficient β from a regression of changes in the spot exchange rate on the forward discount should equal one. Yet a large body of empirical work—spanning four decades, dozens of currencies, and many identification strategies—frequently reports β estimates that are not merely below one but *negative*. This anomaly is known as the **forward premium puzzle** (or, equivalently, the forward discount puzzle). If covered interest parity holds, the result is equivalent to a rejection of uncovered interest parity: currencies with higher interest rates tend to appreciate rather than depreciate.

The puzzle has attracted persistent research activity, with proposed explanations including infrequent portfolio decisions, investor overconfidence, omitted variables, sentiment, sovereign default risk, order flow effects, rare disasters, and monetary policy changes. Despite decades of individual studies, no prior work had synthesized the entire empirical literature quantitatively. This paper fills that gap with a large-scale meta-analysis—collecting 3,643 estimates of β from 91 published studies—to quantify the true magnitude of the puzzle after correcting for publication bias and study-design heterogeneity.

1.2 Structure of the Paper

1. **Section 1 (Introduction):** Motivates the meta-analysis, surveys proposed explanations, and outlines the paper's four contributions.
2. **Section 2 (Testing Forward Rate Unbiasedness):** Presents the theoretical framework (Equations 1–4) and explains both the levels and differences specifications used in the literature.
3. **Section 3 (Data):** Describes the search strategy (Google Scholar), inclusion criteria, dataset construction (3,643 estimates from 91 studies), and descriptive statistics including box plots and histograms.
4. **Section 4 (Publication Bias):** Tests for publication bias using funnel plots, the FAT-PET and PEESE linear/nonlinear frameworks, and five advanced techniques (WAAP, selection model, endogenous kink, p-uniform*, stem method).
5. **Section 5 (Heterogeneity):** Investigates how 42 study-design variables systematically affect reported β using Bayesian Model Averaging (BMA) and Frequentist Model Averaging (FMA), then computes implied β estimates for different currencies.
6. **Section 6 (Extensions & Robustness):** Separately analyzes level estimates, tests robustness to trimming vs. winsorization, examines developed-country subsample, considers interest rate differential studies, replaces currency dummies with GDP and exchange rate regime, and drops survey-based estimates.
7. **Section 7 (Conclusion):** Summarizes findings and discusses implications for explanations of the puzzle.

1.3 Conclusions

After correcting for publication bias using multiple state-of-the-art techniques, the paper finds that the average β from difference regressions is *positive*—ranging from approximately 0.2 to 0.8 depending on the method—far from the naive literature average of -0.6 . The pattern in the

literature is consistent with **confirmation bias**: a tendency to publish results that corroborate the seminal negative-coefficient finding of Fama (1984).

Using Bayesian model averaging to account for heterogeneity across 42 study-design variables and then constructing implied β estimates:

- For **developed-economy currencies**, the implied β lies in the interval (0.23, 0.45)—positive but below one.
- For **emerging-economy currencies**, the implied β lies in (0.95, 1.16)—essentially consistent with the theoretical prediction of $\beta = 1$.
- For specific currencies, negative β estimates persist for the **euro** (−0.79 to −0.57), **Japanese yen** (−0.47 to −0.25), and (marginally) the **Swiss franc** (−0.14 to 0.08).

The results cast doubt on the time-varying risk premium explanation (since emerging currencies with larger and more volatile risk premia show *less* puzzle), and support the role of publication bias, transaction costs/attention, and estimation misspecification.

1.4 Key Takeaways

1. **Publication bias is substantial.** The naive mean β of −0.6 in the literature is heavily distorted by confirmation bias favoring negative estimates. After correction, the mean β is positive (0.2–0.8).
2. **The puzzle is currency-dependent.** Emerging-economy currencies show essentially no puzzle ($\beta \approx 1$). Among developed currencies, the euro, yen, and Swiss franc remain genuinely puzzling, while the French franc and Italian lira show above-average β .
3. **Study design matters systematically.** The choice of estimation method (OLS vs. SUR vs. regime-switching), data frequency (weekly vs. monthly), forward rate horizon, and numeraire currency all significantly influence reported β values.
4. **The puzzle has diminished over time.** Reported β estimates exhibit a statistically significant upward trend of approximately 0.03 per year.
5. **Time-varying risk premia are an unlikely explanation.** If volatile risk premia drove the puzzle, emerging-country currencies (with larger, more volatile premia) should show *more* evidence of it—but they show less.

2 Key Points

2.1 Theoretical Framework: Forward Rate Unbiasedness

The paper starts from the standard asset pricing relationship in logarithms:

$$f_{t,t+k} = E_t s_{t+k} + rp_{t+k}, \quad (1)$$

where $f_{t,t+k}$ is the log forward rate for a contract signed at t maturing k periods ahead, s_{t+k} is the log future spot rate, and rp_{t+k} is a risk premium. Under risk neutrality ($rp_{t+k} = 0$) and rational expectations, one can substitute realized spot rates for expected values and estimate:

$$s_{t+k} = \alpha + \beta \cdot f_{t,t+k} + \nu_t. \quad (2)$$

In practice, most researchers subtract the current spot rate s_t from both sides to obtain the **differences specification**:

$$s_{t+k} - s_t = \alpha + \beta \cdot (f_{t,t+k} - s_t) + \nu_t, \quad (3)$$

where the left-hand side is currency depreciation and the right-hand side is the forward discount. The null hypothesis for forward rate unbiasedness is $\alpha = 0$, $\beta = 1$, and ν_t serially uncorrelated. The literature focuses almost exclusively on β because it determines (i) the profitability of carry trades, (ii) the forecasting power of forward rates, and (iii) a negative β alone suffices to reject the hypothesis. Under covered interest parity, the forward discount equals the interest rate differential $i_{t,t+k} - i_{t,t+k}^*$, linking tests of forward rate unbiasedness to tests of uncovered interest parity (see [Key Terms & Definitions](#)).

2.2 Data Collection and Descriptive Statistics

The authors use Google Scholar to search for studies containing phrases such as “forward rate unbiasedness,” “forward premium puzzle,” and related terms, terminating the search on July 31, 2018. Studies must be published in English, estimate β from Equation (2) or (3), and report standard errors or statistics from which standard errors can be computed.

This yields **3,643 estimates from 91 studies**—one of the largest meta-analyses in economics and finance. After coding, the dataset also contains 42 context variables capturing country scope, data characteristics, estimation methods, regimes, data sources, and publication characteristics, totaling over 150,000 manually collected data points. Estimates and standard errors are winsorized at the 5% level to limit the influence of extreme outliers.

Key descriptive findings include:

- The overall mean β across all 3,643 estimates is -0.34 ; for the 2,989 difference estimates it is -0.60 .
- The histogram of difference estimates is bimodal (peaks near 0 and 1) with a long left tail, suggesting over-representation of negative estimates consistent with confirmation bias.
- Only 654 estimates come from the levels specification (Equation 2), of which only 5.4% are negative. Because levels estimates are likely contaminated by spurious regression (unit roots), the remainder of the analysis focuses on the 2,989 difference estimates.
- The mean β for **advanced-economy currencies** is -0.51 (significantly < 0), while for **emerging-economy currencies** it is $+0.36$ (significantly > 0)—a striking divergence that the paper investigates further.

These patterns are visible in box plots across studies and currencies (Figures 2 and 3 in the paper), which show enormous within- and across-study variation.

2.3 Visual Evidence for Publication Bias: Funnel Plots

The paper presents funnel plots—scatter plots of estimates against their precision (inverse standard error)—as a visual diagnostic for publication bias. In the absence of bias, observations should form a symmetrical inverted funnel around the true mean effect, with more precise estimates clustered tightly and less precise ones dispersed symmetrically.

Two observations emerge from the funnel plots:

1. **Level estimates** do not form a funnel at all; they cluster tightly around $\beta = 1$ regardless of precision, consistent with spurious regression attracting estimates toward unity.
2. **Difference estimates** form an asymmetric funnel. The most precise estimates lie near zero or are mildly positive, but many imprecise positive estimates appear to be missing. This asymmetry is consistent with publication bias that selectively suppresses positive (non-puzzling) estimates, contributing to the observed mean of -0.6 .

The funnel for difference estimates also shows two peaks (at 0 and 1), suggesting heterogeneity in addition to publication bias, which the paper addresses in [Key Point 2.5](#) and [Key Point 2.6](#).

2.4 Publication Bias: Linear and Nonlinear Tests

The paper’s main identification assumption is that, absent publication bias, estimates and their standard errors should be statistically independent. If researchers (intentionally or not) select for statistical significance, a correlation arises—the **Lombard effect** analogy: just as speakers raise their voices in noisy environments without realizing it, researchers try harder to find large estimates when standard errors are large.

FAT-PET (Linear test). The baseline specification divides the meta-regression by the standard error to correct for heteroskedasticity:

$$t_{i,j} = \gamma_1 + \gamma_0 \cdot \frac{1}{SE(\beta_{i,j})} + \omega_i, \quad (4)$$

where $t_{i,j}$ is the t -statistic of estimate i in study j , γ_1 captures publication bias severity, and γ_0 is the bias-corrected mean β (the effect at maximum precision). With study-level fixed effects (FE), the corrected mean β is approximately **0.61**, and publication bias (γ_1) is strongly negative (-2.06). An instrumental variable (IV) specification using $1/\sqrt{n}$ as an instrument for SE yields a smaller corrected mean of about 0.33, still positive and significant.

PEESE (Nonlinear test). The linear assumption may break down at very small or very large standard errors. The PEESE specification replaces SE with SE^2 :

$$t_{i,j} = \gamma_1 \cdot SE(\beta_{i,j}) + \gamma_0 \cdot \frac{1}{SE(\beta_{i,j})} + \xi_i. \quad (5)$$

Results are similar: the corrected mean β is 0.55–0.62 for FE and WLS, and about 0.27 for IV.

Advanced techniques (Panel C). Five recently developed methods are applied:

- **WAAP** (Ioannidis et al., 2017): uses only estimates with statistical power above 80%. Result: $\hat{\beta} = 0.29$.
- **Endogenous kink** (Bom & Rachinger, 2019): estimates a precision threshold below which publication bias operates. Result: $\hat{\beta} = 0.31$.

- **Selection model** (Andrews & Kasy, 2019): models publication probability as a function of p-value cut-offs. Result: $\hat{\beta} = 0.22$.
- **p-uniform*** (van Aert & van Assen, 2021): exploits the principle that p-values are uniformly distributed at the true effect. Result: $\hat{\beta} = 0.78$.
- **Stem method** (Furukawa, 2020): focuses on the most precise fraction of estimates. Result: $\hat{\beta} = 0.79$.

All five methods yield a *positive* corrected mean β , in stark contrast to the naive average of -0.6 . The range (0.22 to 0.79) reflects differences in assumptions, but the qualitative conclusion is unanimous: the literature is substantially distorted by confirmation bias.

2.5 Heterogeneity Analysis via Bayesian Model Averaging

To understand *why* β estimates vary across studies, the authors regress reported β values on 42 study-design variables capturing country scope, data characteristics, estimation methods, regimes, data sources, and publication characteristics:

$$\beta_{ij} = \gamma_0 + \gamma_1 \cdot SE(\beta_{ij}) + \sum_{l=1}^{42} \gamma_l \cdot X_{l,ij} + \zeta_{ij}. \quad (6)$$

Because including all 42 variables simultaneously causes severe model uncertainty and imprecision, the paper uses **Bayesian Model Averaging (BMA)**, which estimates millions of regressions with different subsets of variables and weights models by their posterior model probability (PMP). The baseline uses the unit information prior and the dilution prior (to account for collinearity). Frequentist Model Averaging (FMA) with Mallows' criteria serves as a robustness check.

BMA reports for each variable its **posterior inclusion probability (PIP)**—the sum of PMPs of all models including that variable. Following Jeffreys (1961), $PIP > 0.99$ is “decisive,” 0.95 – 0.99 is “strong,” 0.75 – 0.95 is “positive,” 0.50 – 0.75 is “weak,” and < 0.50 is “irrelevant.”

Twenty variables have $PIP > 0.50$, and 18 have $PIP > 0.75$. The standard error (proxy for publication bias) retains a PIP of 0.95 even after controlling for all 42 variables, confirming the publication bias finding from [Key Point 2.4](#). Key results by category are described in [Key Point 2.6](#) through [Key Point 2.9](#).

2.6 Country Scope: Developed vs. Emerging and Individual Currencies

The choice of currency has the strongest impact on reported β . The BMA results show:

- **Emerging-economy currencies** ($PIP = 1.00$, posterior mean = $+1.44$): estimates are decisively larger than the reference category (mixed samples). This corroborates Frankel & Poonawala (2010).
- **Euro** ($PIP = 1.00$, posterior mean = -1.62): the most negative individual-currency effect.
- **Japanese yen** ($PIP = 1.00$, posterior mean = -0.66): strongly below the developed-country average.
- **French franc** ($PIP = 0.94$, posterior mean = $+0.78$) and **Italian lira** ($PIP = 0.94$, posterior mean = $+0.85$): notably above-average for developed currencies.
- **Advanced currencies** as a group ($PIP = 0.92$, posterior mean = $+0.28$): positive but moderate.

- **British pound and Swiss franc:** low PIP (< 0.20), meaning their individual effects are not robustly distinguishable from the reference.
- **German mark as numeraire** (PIP = 0.95, posterior mean = +0.50): using the mark instead of the USD as the base currency raises β by about 0.5.

The pattern—less risky, more developed currencies showing more puzzle—is *opposite* to what the time-varying risk premium hypothesis would predict. If volatile risk premia were the explanation, riskier currencies should exhibit more bias, not less. This supports the conclusion, shared with Frankel & Poonawala (2010), that time-varying risk premia are not a plausible explanation of the forward premium puzzle.

2.7 Data and Estimation Method Effects

Data characteristics.

- **Weekly frequency** (PIP = 1.00, posterior mean = +0.80): estimates from weekly data are substantially larger than those from other frequencies.
- **One-month horizon** (PIP = 0.88, posterior mean = -0.26): short-horizon forward contracts yield smaller (more negative) estimates, consistent with the UIP literature finding that long-term rates better explain spot rate movements.
- **Time series length** (PIP = 0.93, posterior mean = +0.12): longer samples produce larger β estimates.

Estimation methods.

- **OLS** (PIP = 1.00, posterior mean = -0.57): OLS estimates are substantially lower than those from methods that address endogeneity or cross-sectional dependence.
- **Regime-switching models** (PIP = 0.99, posterior mean = -0.94): produce the smallest estimates on average.
- **Fixed-effects regression** (PIP = 0.88, posterior mean = -0.78) and **SUR** (PIP = 0.75, posterior mean = -0.37): intermediate effects, with SUR estimates being closer to zero (less negative) than OLS, consistent with Fama (1984).

These findings imply that methodological choices matter greatly. More sophisticated techniques (IV, SUR, panel methods) that account for endogeneity tend to yield larger (less puzzling) estimates.

2.8 Regime Effects and Data Sources

Regimes.

- **Large differential** (PIP = 1.00, posterior mean = +2.79): when interest rate or money growth differentials are large, β is dramatically higher. This is consistent with Baillie & Kilic (2006) and Baillie & Chang (2011), who find that UIP is more likely to hold when differentials are large—suggesting that transaction costs or investor attention play a role in the puzzle.
- **Large positive premium** (PIP = 0.55, posterior mean = -0.36): weak evidence that a positive forward premium is associated with slightly smaller β .

Data sources. Datastream-sourced estimates are associated with 0.4 smaller β (PIP = 0.99), which contrasts with the hypothesis that public databases should reduce measurement error and attenuation bias.

Publication characteristics. The year of the first draft has decisive importance (PIP = 1.00, posterior mean = +0.03): β estimates increase by about 0.03 per year, corroborating the upward trend visible in Figure 1. Journal quality (impact factor, citations) does not significantly affect reported β after controlling for other design features.

2.9 Implied β Estimates

The paper constructs “implied” β values by combining BMA coefficients with chosen values for each variable. Three strategies are used:

1. The authors’ preferred methodology (plugging sample maxima for desirable features such as large samples, long horizons, and sophisticated estimation; sample minima for undesirable features such as OLS and short horizons; and sample means elsewhere).
2. The methodology of Frankel & Poonawala (2010).
3. The methodology of Breedon et al. (2016).

The key results across the three strategies:

Currency	Authors’ preferred	F&P (2010)	Breedon et al. (2016)
Advanced currencies	0.31	0.45	0.23
Emerging currencies	0.95	1.16	0.95
Euro	−0.70	−0.57	−0.79
German mark	0.14	0.32	0.10
French franc	0.46	0.63	0.41
British pound	0.07	0.16	−0.06
Italian lira	0.46	0.65	0.43
Japanese yen	−0.35	−0.25	−0.47
Swiss franc	0.06	0.08	−0.14

For emerging economies, the implied β is essentially 1—no puzzle. For developed economies as a whole, β is in (0.23, 0.45): below one but *positive*, sharply contradicting the conventional wisdom of negative β . Persistent negative estimates remain for the euro, yen, and (marginally) Swiss franc, indicating that publication bias alone cannot fully resolve the puzzle for all currencies.

2.10 Extensions and Robustness

The paper presents several extensions that reinforce the baseline findings:

Level estimates. The 654 level estimates cluster tightly around $\beta = 1$ with little evidence of publication bias. All advanced correction techniques yield corrected means near 1. However, the concentration is likely driven by spurious regression in the presence of unit roots, which is why the paper excludes level estimates from its main analysis.

Trimming vs. winsorization. Replacing 5% winsorization with conservative trimming (dropping $|\beta| > 6$ and 3 influential standard errors identified via DFbeta) produces nearly identical results in both publication bias tests and BMA.

Developed-country subsample. Running publication bias tests solely on developed-country estimates still reveals substantial negative publication bias. Corrected mean β ranges from 0.3 to 0.9 (with exceptions for the weak IV instrument and the non-converging selection model).

Interest rate differentials. A supplementary sample of 199 estimates using interest rate differentials (instead of forward rates) shows little evidence of publication bias. The corrected mean β is positive but small, suggesting either a violation of covered interest parity or insufficient data for a definitive conclusion.

Country-level variables. Replacing individual currency dummies with GDP per capita and exchange rate regime confirms that higher GDP per capita is associated with smaller (more puzzling) β . The exchange rate regime itself has no significant effect. Main results are otherwise unchanged.

Excluding survey data. Dropping the few survey-based estimates does not alter the findings.

3 Logical Chains

3.1 From Publication Bias to a Less Puzzling Puzzle

Step 1. Theoretical prediction: Under risk neutrality and rational expectations, $\beta = 1$ in the forward rate unbiasedness regression (Equation 3).

Explanation: This is the null hypothesis that the forward discount is an unbiased predictor of depreciation, derived from standard asset pricing (Key Point 2.1).

Step 2. Empirical observation: The naive mean of reported β in the literature is -0.6 , and 58.4% of difference estimates are negative.

Explanation: Decades of empirical studies have reported negative coefficients, establishing the “forward premium puzzle” as a stylized fact (Key Point 2.2).

Step 3. Detection of asymmetry: Funnel plots reveal that the distribution of estimates is asymmetric—many imprecise positive estimates appear to be missing.

Explanation: If there were no publication selection, imprecise estimates should be symmetrically dispersed around the true mean. The observed asymmetry is consistent with selective suppression of positive (non-puzzling) results (Key Point 2.3).

Step 4. Quantification of publication bias: Both linear (FAT-PET) and nonlinear (PEESE) meta-regression tests detect a statistically significant negative correlation between estimates and standard errors.

Explanation: This correlation arises from the Lombard effect: researchers expend more effort to find large (negative) estimates when noise is high. The bias coefficient γ_1 is significantly negative in all specifications (Key Point 2.4).

Step 5. Corrected mean is positive: After correcting for this bias, all methods yield a mean β between 0.2 and 0.8 for the average currency.

Explanation: The bias-correction techniques extrapolate to the effect at maximum precision (zero standard error), effectively removing the distortion. Five independent advanced methods confirm this finding (Key Point 2.4).

Step 6. Heterogeneity explains remaining variation: BMA shows that currency type, estimation method, data frequency, and regime variables systematically drive the dispersion of β estimates.

Explanation: The standard error retains $\text{PIP} = 0.95$ even after controlling for 42 variables, meaning publication bias persists as an independent source of distortion beyond heterogeneity (Key Point 2.5).

Conclusion. The forward premium puzzle is substantially less puzzling than previously thought. For emerging economies, the implied $\beta \approx 1$ —no puzzle at all. For developed economies, $\beta \in (0.23, 0.45)$ —still below 1 but positive, not negative.

Explanation: The combined effect of correcting for publication bias and accounting for study-design heterogeneity transforms the headline finding from a deeply negative β to a moderately positive one (Key Point 2.9).

3.2 Why Time-Varying Risk Premia Fail as an Explanation

Step 1. The risk premium hypothesis: Fama (1984) proposed that a highly variable rational expectations risk premium could explain $\beta < 1$ and even $\beta < 0$.

Explanation: If the risk premium rp_{t+k} in Equation (1) varies over time and is correlated with the forward discount, the OLS estimate of β will be biased away from 1.

Step 2. Implication for cross-country variation: If time-varying risk premia drive the puzzle, currencies with larger and more volatile risk premia should exhibit more forward bias (lower β).

Explanation: Emerging-economy currencies and riskier developed-country currencies typically carry larger risk premia and exhibit more premium volatility.

Step 3. Empirical finding is opposite: The meta-analysis shows that emerging-economy currencies have $\beta \approx 1$ (no puzzle), while safe-haven developed-country currencies (yen, Swiss franc, euro) have the most negative β .

Explanation: BMA decisively confirms this pattern (PIP = 1.00 for the emerging-currency variable, with a large positive posterior mean of +1.44). Among developed currencies, the French franc and Italian lira—historically riskier—show *less* puzzle ([Key Point 2.6](#)).

Conclusion. Time-varying risk premia cannot plausibly explain the forward premium puzzle. The cross-country pattern is the reverse of what the hypothesis predicts.

Explanation: This result aligns with Frankel & Poonawala (2010) and provides more robust evidence from a much larger sample.

3.3 Transaction Costs and Large Differentials

Step 1. Observation: Periods with large interest rate or money growth differentials are associated with dramatically higher β (BMA posterior mean = +2.79, PIP = 1.00).

Explanation: Large differentials represent large potential profits from carry trades ([Key Point 2.8](#)).

Step 2. Interpretation: When potential profits are large, they exceed transaction costs and attract investor attention, leading to arbitrage that pushes β toward 1.

Explanation: Small profit opportunities may not be worth exploiting after accounting for bid-ask spreads, information costs, and limited attention.

Conclusion. Transaction costs and limited attention plausibly contribute to the forward premium puzzle. The anomaly is smaller precisely when the stakes are high enough to motivate arbitrage.

Explanation: This is consistent with the view that the puzzle exists in the “noise band” where transaction costs prevent full arbitrage.

4 Key Terms & Definitions

Posterior Inclusion Probability (PIP)

Bayesian Model Averaging (BMA)

A technique for addressing model uncertainty by estimating many regressions with different subsets of explanatory variables, weighting each model by its posterior model probability (which balances fit and complexity). The reported posterior mean for each variable is a weighted average across all models. The paper uses BMA with a unit information prior and dilution prior as its baseline, with BRIC and hyper-g priors as robustness checks.

Confirmation bias (in publication)

A form of publication bias in which results consistent with a prior influential finding (here, Fama 1984's negative β) are preferentially selected for reporting and publication, while contradictory results are suppressed. The paper identifies this as a key driver of the negative mean β in the literature.

Covered interest parity (CIP)

The no-arbitrage condition equating the forward discount to the interest rate differential: $f_{t,t+k} - s_t = i_{t,t+k} - i_{t,t+k}^*$. When CIP holds, testing forward rate unbiasedness is equivalent to testing uncovered interest parity.

Differences specification

The regression $s_{t+k} - s_t = \alpha + \beta(f_{t,t+k} - s_t) + \nu_t$, where the left-hand side is depreciation and the right-hand side is the forward discount. This is the dominant specification in the literature (2,989 of 3,643 estimates) and the paper's primary focus.

Dilution prior

A prior used in BMA (George, 2010) that re-weights each model by the determinant of the correlation matrix of included variables, penalizing models with highly collinear regressors.

FAT-PET

Funnel Asymmetry Test (FAT) and Precision-Effect Test (PET). FAT tests whether publication bias exists (is $\gamma_1 \neq 0$?); PET estimates the mean effect corrected for bias (γ_0). Both are obtained from the same regression of t -statistics on inverse standard errors.

Forward premium puzzle

The empirical regularity that β in the differences specification is typically estimated to be well below 1 and often negative, implying that currencies with forward premia (higher interest rates) tend to appreciate rather than depreciate. Equivalently: the forward discount is a biased predictor of future depreciation.

Frequentist Model Averaging (FMA)

An alternative to BMA using Mallows' criteria as model weights. In the paper, FMA uses the orthogonalized covariate space approach of Amini & Parmeter (2012) because estimating all 2^{42} models is infeasible.

Funnel plot

A scatter plot of estimates against their precision ($1/SE$). In the absence of publication bias, the plot should resemble a symmetrical inverted funnel. Asymmetry suggests selective reporting.

Levels specification

The regression $s_{t+k} = \alpha + \beta f_{t,t+k} + \nu_t$. Problematic due to likely non-stationarity of both variables, which can produce spurious estimates near 1.

Lombard effect (analogy)

An analogy for publication bias drawn from psychoacoustics, where speakers involuntarily raise their voices in noisy environments. Similarly, researchers may unconsciously search harder for significant results when data are noisy (standard errors are large), producing a correlation between estimates and standard errors even without deliberate manipulation.

PEESE

Precision-Effect Estimate with Standard Error. A nonlinear extension of FAT-PET that models publication bias as proportional to SE^2 rather than SE , addressing the breakdown of the linear assumption at extreme values of precision.

Posterior Inclusion Probability (PIP)

In BMA, the probability that a given variable belongs in the true model, computed as the sum of posterior model probabilities of all models including that variable. $PIP > 0.75$ is considered at least “positive” evidence.

Publication bias / Selective reporting

The tendency for published results to be a non-random subset of all results obtained by researchers, typically biased toward statistical significance or toward results consistent with prior expectations. The paper uses the term “publication bias” for consistency with earlier literature, while noting that the phenomenon extends beyond journals to working papers.

Risk premium (rp_{t+k})

The compensation demanded by investors for holding a currency perceived as risky. A time-varying risk premium is one of the classical proposed explanations for $\beta \neq 1$, but the paper’s cross-country evidence argues against it.

Uncovered interest parity (UIP)

The hypothesis that the expected change in the exchange rate equals the interest rate differential between two countries. Under CIP, UIP is equivalent to forward rate unbiasedness.

WAAP

Weighted Average of Adequately Powered estimates (Ioannidis et al., 2017). A bias-correction method that assigns non-zero weight only to estimates with statistical power above 80%, thereby discarding the estimates most susceptible to publication bias.

Winsorization

A method of handling outliers by replacing values beyond specified percentiles with the values at those percentiles. The paper winsorizes at the 5% level to prevent extreme estimates (some exceeding 50 in absolute value) from driving results.